Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: An Open-Label, Multi-Drug, Biomarker-Directed, Multi-Centre Phase II Umbrella Study in Patients With Non-Small Cell Lung Cancer, Who Progressed on an Anti-PD-1/PD-L1 Containing Therapy (HUDSON) **Brief Study Title:** Phase II Umbrella Study of Novel Anti-cancer Agents in Participants With NSCLC Who Progressed on an Anti-PD-1/PD-L1 Containing Therapy **Short Title/ Acronym:** HUDSON Study **Trial Status:** Active, not recruiting **Protocol Number:** D6185C00001 **NCT Number:** NCT03334617 **Sponsor/ Company Name:** AstraZeneca **Investigational Product:** Durvalumab (Imfinzi) in combination with ceralasertib (AZD6738), olaparib (Lynparza), oleclumab (MEDI9447), danvatirsen, cediranib (ATC Code: L01EK02), or trastuzumab deruxtecan (Enhertu/T-DXd) **Phase of Development:** Phase II **Medical Monitor:** AstraZeneca Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the objective response rate (ORR) of multiple novel anti-cancer agents combined with durvalumab in participants with metastatic non-small cell lung cancer (NSCLC) who have progressed on an anti-PD-1/PD-L1 containing therapy. **Secondary Objectives:** To evaluate progression-free survival (PFS), overall survival (OS), and the frequency and severity of adverse events (AEs) for the different treatment arms.

2.2 Background & Rationale There is currently no established therapy for participants who have received immune checkpoint inhibitors and platinum-doublet therapies, and novel treatments are urgently needed. Resistance to immune checkpoint blockade is complex, presenting as either primary or acquired resistance. The HUDSON study was designed to build a comprehensive understanding of key features in patients and their tumors associated with disease progression. By utilizing a biomarker-directed umbrella design, the study investigates rational combination therapies targeting proposed mechanisms of immunosuppression with the aim of reinvigorating immune-mediated antitumor activity in the post-ICI setting.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Modular Umbrella Study) **Control Type:** Single-arm per module/cohort **Blinding:** Open-label **Randomization:** N/A (Participants are allocated to modules based on biomarker selection [e.g., ATM, STK11, HRR, HER2] or based on clinical phenotype [primary vs. acquired resistance to prior immunotherapy]).

3.2 Planned Enrollment & Duration Target N: 528 evaluable participants **Number of Sites:** Multi-center (Global) **Estimated Study Start:** 18-Dec-2017 **Estimated Primary Completion:** 13-Sep-2024 *(Note: Estimated overall study completion is Sep 2026)*

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the time of signing the informed consent form. 2. Histologically or cytologically confirmed metastatic or locally advanced and recurrent non-small-cell lung cancer (NSCLC) which is progressing. 3. Received prior anti-programmed cell death protein-1 (PD-1)/programmed death-ligand 1 (PD-L1) containing therapy and a platinum-doublet regimen (separately or in combination). 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1, and a minimum life expectancy of 12 weeks.

4.2 Exclusion Criteria 1. Tumour samples with targetable alterations in EGFR, ALK, ROS1, BRAF, MET, or RET at initial diagnosis. 2. Active or prior documented autoimmune or inflammatory disorders. 3. Active infection including tuberculosis, hepatitis B, hepatitis C, or human immunodeficiency virus (HIV).

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Modular design where all participants receive a backbone of Durvalumab (e.g., 1500 mg Q4W or 1120 mg Q3W) combined with a targeted agent. Assignments include, but are not limited to: oral ceralasertib (240 mg BD), oral olaparib (300 mg BD), IV oleclumab (3000 mg Q2W/Q4W), or IV T-DXd (5.4 mg/kg Q3W) depending on the cohort. **Route of Administration:** Intravenous (IV) for durvalumab, T-DXd, and oleclumab; Oral (PO) for ceralasertib, olaparib, and cediranib (ATC Code: L01EK02). **Duration of Treatment:** Cycles continue until objective radiological disease progression, unacceptable toxicity, or other discontinuation criteria are met.

5.2 Concomitant & Prohibited Therapy Permitted: Bisphosphonates or receptor activator of nuclear factor kappa-B ligand (RANKL) inhibitors for the treatment of bone metastases. **Prohibited:** Any concurrent chemotherapy, immunotherapy, biologic, or hormonal therapy for cancer treatment.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biomarker Profiling (ATM, HER2, STK11, HRR, etc.), Safety Labs
Baseline	Day 0	Module Assignment, First Dose of targeted combination therapy
Interim	Every 3 to 4 Weeks	Safety Labs, Efficacy Assessment (Radiological Imaging per RECIST 1.1), AE monitoring, IP Administration
End of Study	At Progression	Final Vital Signs, Exit Interview, IP Accountability, Transition to Survival Follow-up

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR). **Measurement Method:** Radiological imaging evaluated per RECIST 1.1 criteria (investigator/central review).

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) * Overall Survival (OS) * Frequency and severity of Treatment-Emergent Adverse Events (TEAEs)

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection at each visit; specific monitoring for drug-related AEs associated with combined ICIs and targeted therapies (e.g., pneumonitis, pulmonary embolism, anemia). **Serious Adverse Events (SAEs):** Reported continuously per standard oncology clinical trial timelines (typically 24 hours). **Data Monitoring Committee (DMC):** An independent group of scientists who monitor the safety and scientific integrity of the clinical trial, recommending protocol adjustments or module closures if necessary.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) approach for efficacy endpoints, segregated into biomarker-matched and clinically-defined non-matched resistance cohorts (primary vs. acquired resistance). **Sample Size Justification:** Modular expansion targeting 528 participants, powered to detect early efficacy signals across multiple distinct investigational treatment arms. **Handling of Missing Data:** Standard survival analysis methodologies (e.g., Kaplan-Meier estimates); censoring applied for PFS and OS endpoints.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to ensure accurate biomarker testing, treatment assignment compliance, and robust safety reporting. **Audit Trail:** Electronic Case Report Form (eCRF) locking and centralized data query resolution.

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11. Study Identifiers & Governance

Primary ID: D6185C00001 **Registry Number:** NCT03334617 **Posting ID:** 164590701970941894 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** HUDSON **Official Regulatory Title:** An Open-Label, Multi-Drug, Biomarker-Directed, Multi-Centre Phase II Umbrella Study in Patients With Non-Small Cell Lung Cancer, Who Progressed on an Anti-PD-1/PD-L1 Containing Therapy (HUDSON)

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12. Clinical Framework (Oncology Specific)

Primary Condition: Non-Small Cell Lung Cancer (NSCLC) / Metastatic Non Small Cell Lung Cancer **Preferred UMLS Name:** Non-Small Cell Lung Carcinoma **SNOMED ID:** 457721000124104 **MeSH Code:** D016938 **Diagnosis Threshold:** Metastatic or locally advanced disease progressing on prior immunotherapy (PD-1/PD-L1) and platinum-doublet chemotherapy. **Medical Methodology:** Biomarker-Directed Umbrella Strategy **Optimization Target:** Overcoming immune checkpoint blockade resistance

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13. Study Procedures & Methodology

Management Pathway: Centralized tumor profiling $\rightarrow$ Assignment to biomarker-matched cohort $\rightarrow$ Targeted combination regimen. **Education Protocol (HCP):** Investigator training on module-specific dosing, toxicities, and the overarching umbrella protocol. **Education Protocol (Patient):** Informed consent detailing the genomic screening process and specific modular drug assignments. **Quality Audit Mechanism:** Centralized radiological review by independent facility.

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14. Observation & Follow-Up Schedule

Total Duration: Continuous until disease progression, followed by survival tracking. **Onsite Visit Cadence:** Depending on the assigned treatment arm, typically Every 3 Weeks (Q3W) or Every 4 Weeks (Q4W). **Remote Monitoring Frequency:** Periodic survival follow-up contacts after treatment discontinuation. **Checklist**

Review Interval: Prior to the initiation of each new treatment cycle.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]